Number Patterns

Two rules side by side. How are the numbers related?

Start at 0. Rule A: add 3. Rule B: add 6.

Rule A: add 3	**0**	**3**	**6**	**9**	**12**	**15**
Rule B: add 6	**0**	**6**	**12**	**18**	**24**	**30**

Look down each column: 3 and 6, 6 and 12, 9 and 18.
Each B term is 2 times the A term, because 6 is 2 times 3.

Find the rule

5 9 13 17

Each number is 4 more than the one before.
The rule is: add 4.

Pairs

(3, 6) (6, 12) (9, 18)

Write each A term with its B term.
The A term comes first.

Your turn!

Start at 2, add 5: 2 7 12 17 22

For teachers

Goal: make two patterns from two rules and compare them (5.OA.3). Say “Look down the columns: how do the A and B terms compare?” Pairs get graphed in the Coordinate Grid packet.

Number Patterns: Practice

Find the rule. Keep going. Compare the two rules.

Part 1 Keep going Use the rule.

1) Start at 4, add 6: 4 10 16 **22** **28** **34**

2) Start at 3, add 7: 3 10 17 **24** **31** **38**

3) Start at 10, add 5: 10 15 20 **25** **30** **35**

4) Start at 1, add 9: 1 10 19 **28** **37** **46**

5) Start at 8, add 4: 8 12 16 **20** **24** **28**

6) Start at 5, add 11: 5 16 27 **38** **49** **60**

7) Start at 12, add 8: 12 20 28 **36** **44** **52**

8) Start at 0, add 15: 0 15 30 **45** **60** **75**

Part 2 Find the rule No help this time.

9) 7 12 17 22 rule: add **5**

10) 2 10 18 26 rule: add **8**

11) 11 14 17 20 rule: add **3**

12) 6 18 30 42 rule: add **12**

Number Patterns: Practice (continued)

Part 2 Find the rule (continued)

13) 9 16 23 30 rule: add 7

14) 20 45 70 95 rule: add 25

15) 3 6 12 24 rule: times 2

16) 1 3 9 27 rule: times 3

17) 5 10 20 40 rule: times 2

18) 2 20 200 2000 rule: times 10

Part 3 Two rules Fill in the table. Then compare the columns.

19)

Rule A: add 2	0	2	4	6	8	10
Rule B: add 4	0	4	8	12	16	20

Each B term is 2 times the A term

20)

Rule A: add 4	0	4	8	12	16	20
Rule B: add 12	0	12	24	36	48	60

Each B term is 3 times the A term

21)

Rule A: add 5	0	5	10	15	20	25
Rule B: add 10	0	10	20	30	40	50

Each B term is 2 times the A term

Number Patterns: Practice (continued)

Part 3 Two rules (continued)

22)

Rule A: add 3	0	3	6	9	12	15
Rule B: add 9	0	9	18	27	36	45

Each B term is [3] times the A term

Part 4 Make pairs No help this time. Start at 0. Write the next pair.

23) A: add 2, B: add 6: (0, 0) (2, 6) (4, 12) next: [6] and [18]

24) A: add 5, B: add 15: (0, 0) (5, 15) (10, 30) next: [15] and [45]

25) A: add 4, B: add 8: (0, 0) (4, 8) (8, 16) next: [12] and [24]

26) A: add 3, B: add 12: (0, 0) (3, 12) (6, 24) next: [9] and [36]

27) A: add 6, B: add 18: (0, 0) (6, 18) (12, 36) next: [18] and [54]

28) A: add 10, B: add 30: (0, 0) (10, 30) (20, 60) next: [30] and [90]

Part 5 Compare the rules No help this time. Start at 0.

29) A: add 3, B: add 12. Each B term is [4] times the A term

30) A: add 2, B: add 10. Each B term is [5] times the A term

Number Patterns: Practice (continued)

Part 5 Compare the rules (continued)

31) A: add 5, B: add 20. Each B term is **4** times the A term

32) A: add 4, B: add 24. Each B term is **6** times the A term

33) A: add 6, B: add 12. Each B term is **2** times the A term

34) A: add 7, B: add 21. Each B term is **3** times the A term

Part 6 Keep going

35) Start at 11, add 15: 11 26 41 **56** **71** **86**

36) Start at 13, add 9: 13 22 31 **40** **49** **58**

37) Start at 17, add 5: 17 22 27 **32** **37** **42**

38) Start at 9, add 3: 9 12 15 **18** **21** **24**

39) Start at 3, add 9: 3 12 21 **30** **39** **48**

40) Start at 20, add 5: 20 25 30 **35** **40** **45**

41) Start at 19, add 7: 19 26 33 **40** **47** **54**

Number Patterns: Practice (continued)

Part 6 Keep going (continued)

42) **Start at 16, add 14: 16 30 44** 58 72 86

43) **Start at 2, add 4: 2 6 10** 14 18 22

44) **Start at 4, add 8: 4 12 20** 28 36 44

45) **Start at 18, add 9: 18 27 36** 45 54 63

46) **Start at 8, add 9: 8 17 26** 35 44 53

47) **Start at 5, add 9: 5 14 23** 32 41 50

48) **Start at 20, add 6: 20 26 32** 38 44 50

49) **Start at 12, add 3: 12 15 18** 21 24 27

50) **Start at 14, add 7: 14 21 28** 35 42 49

51) **Start at 9, add 14: 9 23 37** 51 65 79

52) **Start at 12, add 6: 12 18 24** 30 36 42

Number Patterns: Practice (continued)

Part 7 Find the rule

53) 17 32 47 62 rule: add **15**

54) 7 32 57 82 rule: add **25**

55) 12 37 62 87 rule: add **25**

56) 5 28 51 74 rule: add **23**

57) 4 8 12 16 rule: add **4**

58) 5 16 27 38 rule: add **11**

59) 28 46 64 82 rule: add **18**

60) 1 25 49 73 rule: add **24**

61) 28 45 62 79 rule: add **17**

62) 2 23 44 65 rule: add **21**

63) 5 30 55 80 rule: add **25**

64) 19 30 41 52 rule: add **11**

Number Patterns: Practice (continued)

Part 7 Find the rule (continued)

65) 5 18 31 44 rule: add 13

66) 12 34 56 78 rule: add 22

67) 26 33 40 47 rule: add 7

68) 6 11 16 21 rule: add 5

69) 15 24 33 42 rule: add 9

70) 23 45 67 89 rule: add 22

Part 8 Remember? Problems from earlier packets.

71) $(12 + 8) \times 3 = 60$

72) $5 + 6 \times 4 = 29$

73) $\frac{1}{5} \div 2 = \frac{1}{10}$

74) $3 \div \frac{1}{5} = 15$

75) $2.5 \times 4 = 10$

76) $8 \div 11 = \frac{8}{11}$

Number Patterns: Practice (continued)

Part 9 Word problems Make a table if it helps.

77) Mia saves $4 a week. Leo saves $8 a week. When Mia has $20, how much does Leo have?

Number sentence **Leo has 2 times as much: 20 × 2 = 40**

Answer **$40**

Draw or show your work

78) A plant grows 3 cm each week. How tall is it after 6 weeks if it started at 5 cm?

Number sentence **5 + 6 × 3 = 23**

Answer **23** cm

Draw or show your work

79) Ben runs 2 laps a day and Sam runs 6. When Ben has run 10 laps, how many has Sam run?

Number sentence **Sam runs 3 times as many: 10 × 3 = 30**

Answer **30** laps

Draw or show your work

80) The pattern is 7, 14, 21, 28. What is the 10th number?

Number sentence **add 7 each time: 10 × 7 = 70**

Answer **70**

Draw or show your work

81) Challenge: Rule A starts at 0 and adds 5. Rule B starts at 0 and adds 20. What is B when A is 35?

Number sentence **B is 4 times A: 4 × 35 = 140**

Answer **140**

Draw or show your work

Check-In: Number Patterns

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Keep going

1) Start at 2, add 9: 2 11 20 **29** **38** **47**

2) Start at 9, add 6: 9 15 21 **27** **33** **39**

Part 2 Find the rule

3) 5 20 35 50 rule: add **15**

4) 4 13 22 31 rule: add **9**

Part 3 Two rules

5)

Rule A: add 2	**0**	**2**	**4**	**6**	**8**	**10**
Rule B: add 8	**0**	**8**	**16**	**24**	**32**	**40**

Each B term is **4** times the A term

Part 4 Compare the rules

6) A: add 5, B: add 25. Each B term is **5** times the A term

7) A: add 3, B: add 18. Each B term is **6** times the A term